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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

II 312 — Past Questions & Materials
by Renance

Course: II 312
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — electricity and magnetism II 312 PQ

Level 300 • First Semester • PastQuestion

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Course Code: PHY 312 F

ELECTRICITY AND MAGNETISM II

Course Title:

Course Unit: 3 UNITS

Session: 2022/2023

Time Allowed: — 2hrs 30 min Date: 20/05/2023

Instruction: Answer question 1, One in Section A and Two in Section B

1. Explain Lorentz force law
2. Write an equation of Lorentz force law and define all the parameters
3. State (i) Maxwell equations for magnetostatics
(ii) | Maxwell equations for electrostatics
and hence explain the difference between magnetic field and electric field.
- (b) State the three fundamental quantities of magnetostatics
4. State the Biot- Savart law for magnetic field
5. Explain Ferromagnetic material and give four (4) examples
6. State four configurations that can be handled by Ampere's law
7. With the magnetic field produced by a Steady current. Use Biot- Savart Law to explain.
8. Write the two equations for Surface an
written as.

d Volume currents that the Biot- Savart law can be

SECTION A

2. (a) A phonograph record of radius R, carrying a uniform surface charge ϕ ,
angular velocity w . Find its magnetic dipole moment.

(b) State the difference between the Boundary condition for electric field and that of magnetic field

is rotating at constant

A circular loop of wire, with radius R , lies in the xy plane, centred at the origin, and carries a current I running counter clockwise as viewed from the positive z axis. Find

(a) What is its magnetic dipole moment?

(b) What is its (approximate) magnetic field at points far from the origin?

(c) Determine the exact solution for B_z on the positive z axis.

(d) Find the magnetic field at the center of the loop (pointing into the page). The field

is shown in the diagram below.

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| Field Region 1 ϕ

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$1 er OE alk Zee me$

57ϕ

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$1 1$

$1 !$

5. A thick slab extending from $z = -a$ to $z = a$ carries a uniform volume current $\mathbf{J} = J\hat{j}$.

(i) Find the magnetic field both inside and outside the slab.